

Equivariant–Decomposed–Residual (EDR) Networks

Structurally-Constrained Neural Corrections for Robot Inverse Dynamics:
Architecture and Mathematical Formulation

Abstract

The Equivariant–Decomposed–Residual (EDR) network is a grey-box model for robot inverse dynamics. It begins from the analytical rigid-body equation of motion and adds a *small, structurally constrained* learnable correction to *each* of its four physical terms — gravity, inertia, Coriolis/centripetal, and friction. The name states the design: corrections are **decomposed** (one network per physics term, hence interpretable), constrained to be **equivariant** with respect to the symmetry of the term they correct (odd-in-velocity friction, velocity-vanishing Coriolis, symmetric inertia, configuration-only gravity), and applied as an additive **residual** on top of nominal physics so the model *starts* at the analytical solution. This report develops the formulation from Lagrangian mechanics, makes the equivariance structure precise, gives the four correction networks with the linear algebra of their constructions, and proves the structural guarantees (exact symmetry, oddness, velocity-vanishing, passivity, and physics-consistent initialisation). It then specifies the smooth capacity-gate curriculum, the composite training objective, the optimisation/regularisation strategy, and the parameter budget. No experimental accuracy numbers are reported.

1 Introduction and Motivation

Model-based control of a manipulator requires the *inverse-dynamics map* from the joint trajectory $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ to the actuator torques $\boldsymbol{\tau}$. The classical route evaluates the rigid-body dynamics (RBD) of the robot’s URDF model with the Recursive Newton–Euler Algorithm (RNEA, e.g. Pinocchio) and adds an identified friction model, producing a nominal torque $\boldsymbol{\tau}_{\text{phys}}$. In practice $\boldsymbol{\tau}_{\text{phys}}$ is systematically biased by *model–reality gaps*:

- inaccurate inertial parameters (link masses, centres of mass, inertia tensors);
- unmodelled cable, gear, and harmonic-drive friction;
- joint flexibility, backlash, and elasticity;
- simplified geometry (CAD $\neq$ as-built hardware).

The residual $\boldsymbol{\tau}_{\text{meas}} - \boldsymbol{\tau}_{\text{phys}}$ is therefore small but *structured*. Three modelling paradigms address it:

1. **Black-box network:** $\hat{\boldsymbol{\tau}} = f_{\theta}(\tilde{\mathbf{x}})$ maps normalised kinematics $\tilde{\mathbf{x}} \in \mathbb{R}^{3n}$ directly to torque. Maximum capacity, no physics; must relearn the entire rigid-body structure from data.
2. **Physics-regularised network:** an unstructured MLP that *receives* the analytical decomposition as additional inputs and/or is *penalised* for departing from $\boldsymbol{\tau}_{\text{phys}}$. Physics *informs* the network but is not enforced; the map remains a free regressor.
3. **EDR (this work):** physics is built into the *architecture*. Each analytical term carries its own small correction network whose output is constrained, by construction, to share that term’s physical symmetry.

EDR is the natural endpoint of injecting physics into a learner: from *loss* and *input* (physics-regularised) to *structure* (EDR). The three letters of the name encode the contribution:

Symbol	Meaning
n	number of active joints ($n = 5$)
$\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \in \mathbb{R}^n$	joint position, velocity, acceleration
$M(\mathbf{q}) \in \mathbb{R}^{n \times n}$	analytical joint-space inertia matrix (SPD)
$C(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$	analytical Coriolis/centripetal matrix
$\mathbf{g}(\mathbf{q}) \in \mathbb{R}^n$	analytical gravity torque, $\mathbf{g} = \partial V / \partial \mathbf{q}$
$V(\mathbf{q})$	gravitational potential energy
$c_{ijk}(\mathbf{q})$	Christoffel symbol of the first kind
$\boldsymbol{\tau}_g, \boldsymbol{\tau}_M, \boldsymbol{\tau}_C, \boldsymbol{\tau}_f \in \mathbb{R}^n$	nominal components $\mathbf{g}$, $M\ddot{\mathbf{q}}$, $C\dot{\mathbf{q}}$, friction
$\boldsymbol{\tau}_{\text{phys}} = \boldsymbol{\tau}_g + \boldsymbol{\tau}_M + \boldsymbol{\tau}_C + \boldsymbol{\tau}_f$	nominal analytical torque
$\hat{\boldsymbol{\tau}}$	EDR-predicted torque
$\delta \mathbf{g}, \delta M, \delta C, \delta \boldsymbol{\tau}_f$	learnable corrections to $\mathbf{g}$, M , C , $\boldsymbol{\tau}_f$
$\tilde{M} = M + \delta M$, $\tilde{C} = C + \delta C$	corrected inertia / Coriolis
$\gamma \in [0, 1]$	smooth capacity gate on the inertia+Coriolis correction
$\text{Sym}(n)$	vector space of symmetric $n \times n$ matrices
$\odot$	element-wise (Hadamard) product

Table 1: Notation.

Equivariant

each correction respects the symmetry group of its physics term (Section 3); the constraints hold exactly for *every* weight configuration, not approximately and not only on the training distribution.

Decomposed

one network per physical term, so the learned correction is interpretable component-by-component (which physics term is wrong, and by how much).

Residual

corrections are additive on top of nominal physics and near-zero at initialisation, so the model starts at the validated analytical solution and only refines it.

The consequences — developed and justified below — are an order-of-magnitude smaller parameter count, exact physical consistency, interpretability, and a strong inductive bias that aids data efficiency and generalisation.

Notation. Table 1 fixes notation: n is the number of active joints ($n = 5$ for the Kikobot), a tilde ($\tilde{\cdot}$) denotes a z -normalised quantity, $\odot$ is the element-wise (Hadamard) product, $\mathbf{e}_i$ is the i -th standard basis vector, and a δ prefix denotes a learnable correction.

2 Lagrangian Mechanics and the Structure to Preserve

EDR’s constraints are not heuristics; they are the exact symmetries of the manipulator equation. We derive them so the corrections can be designed to inherit them.

2.1 The manipulator equation

For a rigid serial chain the Lagrangian is the kinetic minus potential energy,

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \dot{\mathbf{q}}^\top M(\mathbf{q}) \dot{\mathbf{q}} - V(\mathbf{q}), \quad (1)$$

with $M(\mathbf{q}) \in \text{Sym}(n)$ the configuration-dependent inertia matrix. The Euler–Lagrange equations $\frac{d}{dt}(\partial\mathcal{L}/\partial\dot{\mathbf{q}}) - \partial\mathcal{L}/\partial\mathbf{q} = \boldsymbol{\tau}$ give, after expanding $\frac{d}{dt}(M\dot{\mathbf{q}}) = M\ddot{\mathbf{q}} + \dot{M}\dot{\mathbf{q}}$,

$$M(\mathbf{q})\ddot{\mathbf{q}} + \left(\dot{M}(\mathbf{q})\dot{\mathbf{q}} - \frac{1}{2} \nabla_{\mathbf{q}}(\dot{\mathbf{q}}^\top M \dot{\mathbf{q}}) \right) + \nabla_{\mathbf{q}}V(\mathbf{q}) = \boldsymbol{\tau}. \quad (2)$$

Identifying the velocity-quadratic bracket as $C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ and the potential gradient as $\mathbf{g}(\mathbf{q}) = \nabla_{\mathbf{q}}V$, and appending a dissipative friction torque $\boldsymbol{\tau}_f(\dot{\mathbf{q}})$ that is outside the conservative Lagrangian, yields the standard form

$$\boxed{\boldsymbol{\tau} = M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \boldsymbol{\tau}_f(\dot{\mathbf{q}})} \quad (3)$$

with the four physically distinct components $\boldsymbol{\tau}_M = M\ddot{\mathbf{q}}$, $\boldsymbol{\tau}_C = C\dot{\mathbf{q}}$, $\boldsymbol{\tau}_g = \mathbf{g}$, $\boldsymbol{\tau}_f$, summing to $\boldsymbol{\tau}_{\text{phys}}$.

2.2 Christoffel symbols and the inertia–Coriolis link

Writing the Coriolis term componentwise gives the Christoffel symbols of the first kind,

$$C_{ij}(\mathbf{q}, \dot{\mathbf{q}}) = \sum_{k=1}^n c_{ijk}(\mathbf{q}) \dot{q}_k, \quad c_{ijk} = \frac{1}{2} \left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i} \right). \quad (4)$$

Equation (4) shows that C is *not independent* of M : it is fully determined by the spatial derivatives of the inertia matrix. A model that learns M and C independently can violate this coupling; EDR offers a mode (Section 6.4) that preserves it exactly.

2.3 The five structural properties

Equation (3) possesses five properties that any physically valid model must respect.

Property 1 (Inertia symmetry / SPD). $M(\mathbf{q}) = M(\mathbf{q})^\top$ and $\mathbf{x}^\top M(\mathbf{q})\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Property 2 (Gravity depends only on $\mathbf{q}$). $\mathbf{g} = \nabla_{\mathbf{q}}V(\mathbf{q})$ is independent of $\dot{\mathbf{q}}, \ddot{\mathbf{q}}$.

Property 3 (Coriolis vanishes at rest, quadratic in velocity). $C(\mathbf{q}, \mathbf{0})\mathbf{0} = \mathbf{0}$; moreover $\boldsymbol{\tau}_C$ is a homogeneous quadratic form in $\dot{\mathbf{q}}$.

Property 4 (Friction is odd in velocity). $\boldsymbol{\tau}_f(-\dot{\mathbf{q}}) = -\boldsymbol{\tau}_f(\dot{\mathbf{q}})$ (reversing motion reverses the dissipative torque).

Property 5 (Passivity / skew-symmetry). $N(\mathbf{q}, \dot{\mathbf{q}}) := \dot{M} - 2C$ is skew-symmetric: $N = -N^\top$.

Property 5 is the dynamical fingerprint of energy conservation and is a prerequisite for Lyapunov-based control stability. It follows from (4):

Lemma 1. *If C is built from M via (4), then $\dot{M} - 2C$ is skew-symmetric.*

Proof. With $\dot{M}_{ij} = \sum_k (\partial M_{ij} / \partial q_k) \dot{q}_k$ and $2C_{ij} = \sum_k (\partial_k M_{ij} + \partial_j M_{ik} - \partial_i M_{jk}) \dot{q}_k$,

$$N_{ij} = \dot{M}_{ij} - 2C_{ij} = \sum_k (\partial_i M_{jk} - \partial_j M_{ik}) \dot{q}_k.$$

Swapping $i \leftrightarrow j$ gives $N_{ji} = \sum_k (\partial_j M_{ik} - \partial_i M_{jk}) \dot{q}_k = -N_{ij}$. $\square$

EDR’s design principle: *every correction inherits the property of the term it corrects*, so the corrected model satisfies Properties 1–5.

3 Equivariance: the Symmetries Each Correction Preserves

The properties above can be phrased as equivariance under two group actions, which is what the architecture enforces.

Time-reversal ($\mathbb{Z}_2$) parity. Consider the involution $T : \dot{\mathbf{q}} \mapsto -\dot{\mathbf{q}}$ (with $\mathbf{q}, \ddot{\mathbf{q}}$ fixed), the kinematic signature of reversing the direction of motion. A torque term $\mathbf{t}$ is *T-even* if $\mathbf{t}(T\cdot) = \mathbf{t}(\cdot)$ and *T-odd* if $\mathbf{t}(T\cdot) = -\mathbf{t}(\cdot)$. From (3):

$$\underbrace{\tau_M = M\ddot{\mathbf{q}}}_{T\text{-even}}, \quad \underbrace{\tau_C = C\dot{\mathbf{q}}}_{T\text{-even}}, \quad \underbrace{\tau_g = \mathbf{g}(\mathbf{q})}_{T\text{-even}}, \quad \underbrace{\tau_f(\dot{\mathbf{q}})}_{T\text{-odd}}. \quad (5)$$

τ_C is *T-even* because C is odd in $\dot{\mathbf{q}}$ (linear, vanishing at $\dot{\mathbf{q}} = 0$) and $\dot{\mathbf{q}}$ is odd, so their product is even. The conservative terms are *T-even*; only dissipation is *T-odd*. EDR requires each correction to carry the parity of its term: $\delta\mathbf{g}$, $\delta M\ddot{\mathbf{q}}$ are *T-even* by construction (independent of $\dot{\mathbf{q}}$, or multiplied by the even $\ddot{\mathbf{q}}$); $\delta\tau_f$ is forced *T-odd* exactly (Section 6.5); and $\delta C\dot{\mathbf{q}}$ is forced to vanish at $\dot{\mathbf{q}} = \mathbf{0}$ in all modes, and to be exactly the *T-even* quadratic form in the structural mode.

Transpose ($O(n)$ -type) symmetry. The inertia matrix lives in the symmetric subspace $\text{Sym}(n) \subset \mathbb{R}^{n \times n}$, invariant under transposition. The correction δM is constrained to $\text{Sym}(n)$ exactly (Section 6.3), i.e. equivariant under the transpose action $X \mapsto X^\top$ with $\delta M \mapsto \delta M$.

Rest invariance. Both Coriolis and friction must satisfy $\mathbf{t}(\mathbf{q}, \mathbf{0}) = \mathbf{0}$ (no torque at rest, ignoring static gravity). EDR enforces this with a multiplicative velocity factor (Section 6.4, 6.5), which is exact for any network weights.

These are the equivariances meant by the “E” in EDR: they are baked into the maps, so no parameter setting can break them.

4 Data Interface and Normalisation

EDR consumes two tensors per sample, as produced by the data loader.

Kinematic features.

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{q}}^\top, \tilde{\dot{\mathbf{q}}}^\top, \tilde{\ddot{\mathbf{q}}}^\top]^\top \in \mathbb{R}^{3n}, \quad \tilde{\mathbf{q}} = \frac{\mathbf{q} - \boldsymbol{\mu}_q}{\boldsymbol{\sigma}_q}, \quad \tilde{\dot{\mathbf{q}}} = \frac{\dot{\mathbf{q}} - \boldsymbol{\mu}_{\dot{\mathbf{q}}}}{\boldsymbol{\sigma}_{\dot{\mathbf{q}}}}, \quad \tilde{\ddot{\mathbf{q}}} = \frac{\ddot{\mathbf{q}} - \boldsymbol{\mu}_{\ddot{\mathbf{q}}}}{\boldsymbol{\sigma}_{\ddot{\mathbf{q}}}}, \quad (6)$$

with per-joint statistics estimated on the training split.

Decomposed physics. The analytical decomposition is supplied as a $4n$ -vector

$$\mathbf{p} = [\tilde{\tau}_g^\top, \tilde{\tau}_M^\top, \tilde{\tau}_C^\top, \tilde{\tau}_f^\top]^\top \in \mathbb{R}^{4n}, \quad \tilde{\tau}_x = \frac{\tau_x - \boldsymbol{\mu}_\tau/4}{\boldsymbol{\sigma}_\tau}, \quad x \in \{g, M, C, f\}, \quad (7)$$

each component normalised by the *same* torque statistics $(\boldsymbol{\mu}_\tau, \boldsymbol{\sigma}_\tau)$ used for the target, with the mean split evenly across the four additive parts.

Proposition 1 (Additivity in normalised space). *With the target normalised as $\tilde{\boldsymbol{\tau}} = (\boldsymbol{\tau} - \boldsymbol{\mu}_\tau)/\boldsymbol{\sigma}_\tau$, $\sum_x \tilde{\tau}_x = \tilde{\boldsymbol{\tau}}$.*

Proof. $\sum_x \tilde{\tau}_x = (\sum_x \tau_x - 4 \cdot \boldsymbol{\mu}_\tau/4)/\boldsymbol{\sigma}_\tau = (\boldsymbol{\tau} - \boldsymbol{\mu}_\tau)/\boldsymbol{\sigma}_\tau = \tilde{\boldsymbol{\tau}}$, using $\sum_x \tau_x = \boldsymbol{\tau}$. $\square$

Proposition 1 is what lets EDR assemble its prediction directly in normalised coordinates (Section 5) and compare to the normalised target with no re-scaling; the physical prediction is recovered as $\hat{\boldsymbol{\tau}} = \tilde{\boldsymbol{\tau}} \odot \boldsymbol{\sigma}_\tau + \boldsymbol{\mu}_\tau$.

Algorithm 1 (EDR forward pass)**Input:** normalised $\tilde{\mathbf{x}} \rightarrow (\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$; physics $\mathbf{p} \rightarrow (\boldsymbol{\tau}_g, \boldsymbol{\tau}_M, \boldsymbol{\tau}_C, \boldsymbol{\tau}_f)$; gate γ .1. Build features: $\mathbf{q}_{\text{raw}} = \tilde{\mathbf{q}} \odot \boldsymbol{\sigma}_q + \boldsymbol{\mu}_q$; $\mathbf{z}_{\text{cfg}} = [\tilde{\mathbf{q}}, \sin \mathbf{q}_{\text{raw}}, \cos \mathbf{q}_{\text{raw}}]$; velocity products $\{\dot{q}_j \dot{q}_k\}_{j \leq k}$; append physics slices (Section 7).2. $\delta \mathbf{g} = \text{MLP}_g(\mathbf{z}_g)$.3. $\delta M = \text{scatter}_{\text{Sym}}(\text{MLP}_M(\mathbf{z}_M)) \in \text{Sym}(n)$; $\delta M \ddot{\mathbf{q}}$.4. $\delta C \dot{\mathbf{q}} = \dot{\mathbf{q}} \odot \text{MLP}_C(\mathbf{z}_C)$ (or matrix / Christoffel form, Section 6.4).5. $\delta \boldsymbol{\tau}_f = \dot{\mathbf{q}} \odot h_\phi(|\dot{\mathbf{q}}|, \|\dot{\mathbf{q}}\|, \|\boldsymbol{\tau}_f\|)$ (or Stribeck form).6. $\hat{\boldsymbol{\tau}} = (\boldsymbol{\tau}_g + \delta \mathbf{g}) + (\boldsymbol{\tau}_M + \gamma \delta M \ddot{\mathbf{q}}) + (\boldsymbol{\tau}_C + \gamma \delta C \dot{\mathbf{q}}) + (\boldsymbol{\tau}_f + \delta \boldsymbol{\tau}_f)$.**Return** $\hat{\boldsymbol{\tau}}$ (and $\{\delta \mathbf{g}, \delta M, \delta C \dot{\mathbf{q}}, \delta \boldsymbol{\tau}_f\}$ for the regulariser).

5 The EDR Decomposition

EDR corrects each analytical component with its own learned term:

$$\hat{\boldsymbol{\tau}} = \underbrace{(\boldsymbol{\tau}_g + \delta \mathbf{g}(\mathbf{q}))}_{\text{gravity}} + \underbrace{(\boldsymbol{\tau}_M + \gamma \delta M(\mathbf{q}) \ddot{\mathbf{q}})}_{\text{inertia}} + \underbrace{(\boldsymbol{\tau}_C + \gamma \delta C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}})}_{\text{Coriolis}} + \underbrace{(\boldsymbol{\tau}_f + \delta \boldsymbol{\tau}_f(\dot{\mathbf{q}}))}_{\text{friction}} \quad (8)$$

equivalently, a *structured residual* on nominal physics,

$$\hat{\boldsymbol{\tau}} = \boldsymbol{\tau}_{\text{phys}} + \delta \mathbf{g}(\mathbf{q}) + \gamma \delta M(\mathbf{q}) \ddot{\mathbf{q}} + \gamma \delta C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \delta \boldsymbol{\tau}_f(\dot{\mathbf{q}}). \quad (9)$$

All quantities are in normalised space (Section 4). Two model-level mechanisms appear: *near-zero initialisation* so that $\hat{\boldsymbol{\tau}} \approx \boldsymbol{\tau}_{\text{phys}}$ at the start (Section 9), and the *capacity gate* $\gamma \in [0, 1]$, which multiplies only the inertia and Coriolis corrections (the two highest-capacity, $O(n^2)$ terms) and is annealed $0 \rightarrow 1$ over early training (Section 8); $\gamma = 1$ at inference. Algorithm 5 summarises the forward pass.

6 The Four Correction Networks

6.1 Shared building block: the near-zero-initialised MLP

Each correction is produced by a small MLP with L layers, shared activation ϕ (SiLU, $\phi(x) = x \sigma(x)$), Xavier-normal hidden weights, zero hidden biases, and dropout rate p after each hidden activation:

$$\text{MLP}(\mathbf{z}) = W^{(L)} \phi(\dots \phi(W^{(1)} \mathbf{z} + \mathbf{b}^{(1)}) \dots) + \mathbf{b}^{(L)}, \quad W^{(L)} \sim \mathcal{N}(0, \sigma_0^2), \mathbf{b}^{(L)} = \mathbf{0}, \quad (11)$$

with the final layer initialised *near zero*, $\sigma_0 = 10^{-4}$. The reason the final layer is near-zero rather than exactly zero is analysed in Section 9. Optional spectral normalisation caps the Lipschitz constant of the hidden layers (a generalisation regulariser) while leaving the final layer — hence all structural constructions — untouched.

6.2 Gravity correction $\delta \mathbf{g}(\mathbf{q})$

Gravity is the gradient of a configuration potential (Property 2), so $\delta \mathbf{g}$ is a free MLP of configuration features only:

$$\delta \mathbf{g}(\mathbf{q}) = \text{MLP}_g(\tilde{\mathbf{q}}, \sin \mathbf{q}_{\text{raw}}, \cos \mathbf{q}_{\text{raw}}) \in \mathbb{R}^n. \quad (12)$$

The trigonometric augmentation supplies the correct functional basis: gravity torques are sums of products of $\sin / \cos$ of joint angles, so feeding $\sin \mathbf{q}_{\text{raw}}, \cos \mathbf{q}_{\text{raw}}$ removes the need for the MLP to approximate periodic functions from $\tilde{\mathbf{q}}$ alone. No further constraint is needed — $\delta \mathbf{g}$ is a free additive vector, automatically T -even and $\dot{\mathbf{q}}, \ddot{\mathbf{q}}$ -independent, so the corrected gravity $\boldsymbol{\tau}_g + \delta \mathbf{g}$ keeps Property 2.

6.3 Inertia correction $\delta M(\mathbf{q}) \ddot{\mathbf{q}}$

The correction must lie in $\text{Sym}(n)$ (Property 1). The MLP emits the $m := n(n+1)/2$ free coordinates of a symmetric matrix, which are scattered into both triangles.

Symmetric scatter (exact symmetry). Let $\mathcal{S} = \{(i, j) : 1 \leq j \leq i \leq n\}$ index the lower triangle, $|\mathcal{S}| = m$, and define the symmetric basis matrices

$$E^{(i,j)} = \begin{cases} \mathbf{e}_i \mathbf{e}_i^\top, & i = j, \\ \mathbf{e}_i \mathbf{e}_j^\top + \mathbf{e}_j \mathbf{e}_i^\top, & i > j. \end{cases} \quad (13)$$

With MLP output $\mathbf{c}(\mathbf{q}) = \text{MLP}_M(\mathbf{z}_M) \in \mathbb{R}^m$, the correction is

$$\delta M(\mathbf{q}) = \sum_{(i,j) \in \mathcal{S}} c_{(i,j)}(\mathbf{q}) E^{(i,j)}, \quad \implies \quad \delta M = \delta M^\top. \quad (14)$$

The map $\mathbf{c} \mapsto \delta M$ is a linear isomorphism $\mathbb{R}^m \xrightarrow{\sim} \text{Sym}(n)$; equivalently $\text{vec}(\delta M) = D_n \mathbf{c}$ with D_n the $n^2 \times m$ duplication matrix. The torque contribution is the matrix–vector product $\delta M(\mathbf{q}) \ddot{\mathbf{q}}$.

Proposition 2 (Exact symmetry). $\delta M(\mathbf{q}) \in \text{Sym}(n)$ for every $\mathbf{c} \in \mathbb{R}^m$ and every $\mathbf{q}$.

Proof. Each $E^{(i,j)}$ is symmetric, and $\text{Sym}(n)$ is a linear subspace closed under real linear combinations; hence the sum (14) is symmetric. $\square$

Why not a Cholesky ($LL^\top$) factor? A common SPD parametrisation sets $\delta M = L(\mathbf{q})L(\mathbf{q})^\top$ with L lower-triangular. EDR uses the additive scatter (14) as default for two reasons. First, $LL^\top \succeq 0$ forces the correction to be positive-semidefinite, i.e. it can only *increase* effective inertia; but the URDF often *over*-estimates inertia, so the data demand indefinite corrections that also reduce it. The additive scatter spans all of $\text{Sym}(n)$, including indefinite directions. Second, $LL^\top$ creates a gradient deadlock at zero initialisation, formalised next. (When a guaranteed-SPD correction is desired — e.g. for control-theoretic use — EDR offers the $LL^\top$ mode; then $\tilde{M} = M + \delta M \succ 0$ since $M \succ 0$.)

Proposition 3 (Gradient deadlock of exact-zero $LL^\top$). For $\delta M = LL^\top$ with L produced by a linear final layer initialised to exactly zero, the gradient of any loss w.r.t. that layer is zero at initialisation, so L never leaves $\mathbf{0}$. The additive scatter (14) does not suffer this.

Proof. For $Y = LL^\top \ddot{\mathbf{q}}$, $\partial Y / \partial L_{ab} = E_{ab} L^\top \ddot{\mathbf{q}} + L E_{ab}^\top \ddot{\mathbf{q}}$ (with $E_{ab} = \mathbf{e}_a \mathbf{e}_b^\top$), which vanishes at $L = 0$; by the chain rule the final-layer weights producing L receive zero gradient and remain zero. For the scatter, $\partial(\delta M \ddot{\mathbf{q}}) / \partial c_{(i,j)} = E^{(i,j)} \ddot{\mathbf{q}}$, independent of $\mathbf{c}$ and generically non-zero, so gradients flow from the first step. $\square$

Proposition 3 is why the shared MLP uses a *near*-zero ($\sigma_0 = 10^{-4}$) rather than exactly-zero final layer: it keeps the $LL^\top$ mode trainable and yields $\delta M = O(\sigma_0^2) \approx 0$ at init regardless of mode.

6.4 Coriolis correction $\delta C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}}$

The correction must vanish at rest (Property 3). EDR provides three constructions, all enforcing this exactly; they differ in how much of the quadratic/passive structure they additionally guarantee.

(a) Element-wise velocity gating (per-joint vanishing).

$$\delta C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} = \dot{\mathbf{q}} \odot \text{MLP}_C(\mathbf{z}_C) \in \mathbb{R}^n. \quad (15)$$

The Hadamard factor $\dot{\mathbf{q}}$ forces joint i 's correction to vanish whenever $\dot{q}_i = 0$. Because the multiplier is $\dot{\mathbf{q}}$ itself, the constraint $\delta C(\mathbf{q}, \mathbf{0})\mathbf{0} = \mathbf{0}$ holds for *any* MLP output. The feature vector $\mathbf{z}_C$ contains the velocity products $\{\dot{q}_j \dot{q}_k\}_{j \leq k}$ (the upper triangle of $\dot{\mathbf{q}} \dot{\mathbf{q}}^\top$), the natural basis of the quadratic Coriolis form $(\boldsymbol{\tau}_C)_i = \sum_{jk} \Gamma_{ijk}(\mathbf{q}) \dot{q}_j \dot{q}_k$.

(b) Matrix form (whole-vector vanishing, cross-joint coupling).

$$\delta C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} = B(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}}, \quad B = \text{reshape}_{n \times n}(\text{MLP}_C(\mathbf{z}_C)). \quad (16)$$

Since $B\mathbf{0} = \mathbf{0}$ for every B , the correction vanishes iff the *whole* velocity vector is zero, while allowing joint i 's correction to be non-zero when $\dot{q}_i = 0$ provided another joint moves — the physically correct coupling that (15) cannot express.

(c) Structural / Christoffel form (passive by construction). The Coriolis correction is *derived from* δM rather than learned independently, using (4) applied to the correction:

$$\delta c_{ijk}(\mathbf{q}) = \frac{1}{2} \left(\frac{\partial \delta M_{ij}}{\partial q_k} + \frac{\partial \delta M_{ik}}{\partial q_j} - \frac{\partial \delta M_{jk}}{\partial q_i} \right), \quad (\delta C \dot{\mathbf{q}})_i = \sum_{j,k} \delta c_{ijk} \dot{q}_j \dot{q}_k. \quad (17)$$

The Jacobian $\partial \delta M / \partial \mathbf{q}$ is evaluated through the *same* feature map δM consumes (so the two δM call-sites cannot drift), in practice via reverse-mode automatic differentiation vectorised over the batch (`vmap` $\circ$ `jacrev` over the small δM -MLP). This form adds *zero* parameters and yields exact passivity:

Proposition 4 (Exact passivity of the corrected model). *If δC is built from δM via (17), then $\dot{\tilde{M}} - 2\tilde{C}$ is skew-symmetric, where $\tilde{M} = M + \delta M$, $\tilde{C} = C + \delta C$.*

Proof. By Lemma 1 the nominal $\dot{M} - 2C$ is skew. The map $M \mapsto C$ in (4) and the operator $M \mapsto \dot{M} - 2C$ are linear in M ; applying the same construction to δM gives $\delta \dot{M} - 2\delta C$ skew. The sum of skew matrices is skew, so $\dot{\tilde{M}} - 2\tilde{C} = (\dot{M} - 2C) + (\delta \dot{M} - 2\delta C)$ is skew. $\square$

Form (c) thus makes Property 5 hold *exactly*, with no soft penalty, and removes the independent (high-capacity) δC network. Forms (a)/(b) guarantee only velocity-vanishing; passivity for them, if desired, is encouraged by the soft stability loss of Section 11.

6.5 Friction correction $\delta \boldsymbol{\tau}_f(\dot{\mathbf{q}})$

Friction is odd in velocity (Property 4). EDR enforces oddness with the *product trick*: an odd factor times an even function. Define the smooth even magnitude $|\dot{\mathbf{q}}|_\varepsilon = \sqrt{\dot{\mathbf{q}} \odot \dot{\mathbf{q}} + \varepsilon}$ ($\varepsilon = 10^{-6}$), differentiable at $\dot{\mathbf{q}} = \mathbf{0}$ with vanishing derivative there (matching stick-slip behaviour).

(a) MLP form.

$$\delta \boldsymbol{\tau}_f(\dot{\mathbf{q}}) = \dot{\mathbf{q}} \odot h_\phi(|\dot{\mathbf{q}}|_\varepsilon, [|\ddot{\mathbf{q}}|_\varepsilon], [|\boldsymbol{\tau}_f|_\varepsilon]), \quad (18)$$

where h_ϕ is an MLP whose inputs are all even in $\dot{\mathbf{q}}$. Optional channels feed the acceleration magnitude (stiction/transition effects) and the analytical friction magnitude $|\boldsymbol{\tau}_f|_\varepsilon$ (physics conditioning; $\boldsymbol{\tau}_f$ is odd, so $|\boldsymbol{\tau}_f|_\varepsilon$ is even and preserves the parity).

Proposition 5 (Exact oddness). $\delta \boldsymbol{\tau}_f(-\dot{\mathbf{q}}) = -\delta \boldsymbol{\tau}_f(\dot{\mathbf{q}})$ for every ϕ .

Proof. $|\dot{\mathbf{q}}|_\varepsilon = |\dot{\mathbf{q}}|_\varepsilon$ (even argument), so h_ϕ is unchanged; the prefactor $\dot{\mathbf{q}}$ flips sign: $\delta \boldsymbol{\tau}_f(-\dot{\mathbf{q}}) = (-\dot{\mathbf{q}}) \odot h_\phi(|\dot{\mathbf{q}}|_\varepsilon) = -\delta \boldsymbol{\tau}_f(\dot{\mathbf{q}})$. The optional channels $|\ddot{\mathbf{q}}|_\varepsilon, |\boldsymbol{\tau}_f|_\varepsilon$ are even and do not affect the argument. $\square$

(b) Stribeck form. A structured Coulomb + Stribeck + viscous law, with the four per-joint parameters (F_c, F_s, v_s, F_v) emitted by the MLP:

$$\delta\tau_f(\dot{\mathbf{q}}) = \text{sgn}_s(\dot{\mathbf{q}}) \odot \left[F_c + F_s e^{-(|\dot{\mathbf{q}}|_\varepsilon / v_s)^2} \right] + F_v \odot \dot{\mathbf{q}}, \quad \text{sgn}_s(\dot{\mathbf{q}}) = \frac{\dot{\mathbf{q}}}{|\dot{\mathbf{q}}|_\varepsilon}, \quad v_s = \text{softplus}(\cdot) + 10^{-3}. \quad (19)$$

Here sgn_s is exactly odd, the bracket is even, and $F_v \odot \dot{\mathbf{q}}$ is odd, so $\delta\tau_f$ is exactly odd (Proposition 5 extends verbatim) while expressing the Coulomb offset and the Stribeck friction dip that the pure-MLP form cannot. The softplus keeps $v_s > 0$.

7 Input Feature Construction

A single feature-builder assembles physics-informed inputs shared by the four networks.

- **Trigonometric features:** raw angles $\mathbf{q}_{\text{raw}} = \tilde{\mathbf{q}} \odot \boldsymbol{\sigma}_q + \boldsymbol{\mu}_q$ are reconstructed and the configuration block becomes $\mathbf{z}_{\text{cfg}} = [\tilde{\mathbf{q}}, \sin \mathbf{q}_{\text{raw}}, \cos \mathbf{q}_{\text{raw}}] \in \mathbb{R}^{3n}$, the correct basis for the periodic gravity/inertia dependence.
- **Velocity products:** the upper triangle of $\dot{\mathbf{q}}\dot{\mathbf{q}}^\top$, $\{\dot{q}_j\dot{q}_k : j \leq k\} \in \mathbb{R}^{n(n+1)/2}$, the quadratic basis for Coriolis.
- **Physics conditioning:** the matching analytical component is threaded into each network — $\delta\mathbf{g}$ also sees $\tilde{\boldsymbol{\tau}}_g$, $\delta\mathbf{M}$ sees $\tilde{\boldsymbol{\tau}}_M$, $\delta\mathbf{C}$ sees $\tilde{\boldsymbol{\tau}}_C$, friction sees $|\tilde{\boldsymbol{\tau}}_f|_\varepsilon$ (an even channel). This delivers the same analytical information the physics-regularised baseline gets, but to the correct structural sub-module.

With physics conditioning enabled, the per-network input dimensions for $n = 5$ are

$$\begin{aligned} \dim \mathbf{z}_g = \dim \mathbf{z}_M = 3n + n = 20, \quad \dim \mathbf{z}_f = n + n + n = 15, \\ \dim \mathbf{z}_C = \underbrace{3n}_{\tilde{\mathbf{q}}, \sin, \cos} + \underbrace{n}_{\dot{\mathbf{q}}} + \underbrace{\frac{n(n+1)}{2}}_{\dot{q}_j\dot{q}_k} + \underbrace{n}_{\tilde{\boldsymbol{\tau}}_C} = 40. \end{aligned}$$

Without physics conditioning $\dim \mathbf{z}_C = 35$ and $\dim \mathbf{z}_g = \dim \mathbf{z}_M = 15$, $\dim \mathbf{z}_f = 5$.

8 Curriculum Learning: from Two-Phase to a Smooth Gate

The inertia and Coriolis corrections carry $O(n^2)$ effective capacity. If given full strength from the first epoch, the $\delta\mathbf{M}\ddot{\mathbf{q}}$ term has enough freedom to partially absorb the large *gravity* and *friction* residuals, which would corrupt the decomposition’s interpretability. EDR therefore introduces the easy $(\delta\mathbf{g}, \delta\boldsymbol{\tau}_f)$ corrections first and the hard $(\delta\mathbf{M}, \delta\mathbf{C})$ corrections later.

Two-phase curriculum (original). Phase 1 trains only $\delta\mathbf{g}, \delta\boldsymbol{\tau}_f$ ($\delta\mathbf{M}, \delta\mathbf{C}$ frozen); Phase 2 unfreezes all four. The transition is triggered adaptively when the validation RMSE plateaus,

$$\frac{\text{RMSE}_{t-W} - \text{RMSE}_t}{\text{RMSE}_{t-W}} < \theta \implies \text{enter Phase 2}, \quad (20)$$

with smoothing window W , threshold θ , a minimum epoch (noise guard) and a maximum epoch (safety cap). At transition the optimiser unfreezes $\delta\mathbf{M}, \delta\mathbf{C}$, equalises learning rates, and clears stale Adam state.

Smooth capacity gate (current). The hard freeze/unfreeze injects an optimiser discontinuity. EDR replaces it with a scalar gate $\gamma \in [0, 1]$ that multiplies the inertia+Coriolis contribution in (8) and is cosine-ramped from 0 to 1 over the first fraction ρ of the epoch budget E :

$$\gamma(e) = \begin{cases} \frac{1}{2} \left(1 - \cos \frac{\pi(e-1)}{\lceil \rho E \rceil} \right), & 1 \leq e \leq \lceil \rho E \rceil, \\ 1, & e > \lceil \rho E \rceil. \end{cases} \quad (21)$$

All parameters now train from epoch 1 under a single optimiser group: Adam momentum stays continuous, there is no learning-rate jump, and the validation curve descends smoothly over a long horizon. The gate achieves the curriculum’s goal — gravity/friction absorb the easy residuals while $\delta M, \delta C$ are still near-silent — without the discontinuity. At inference $\gamma = 1$.

9 Initialisation at the Physics Baseline

Proposition 6 (Physics-consistent start). *With every final layer initialised to $W^{(L)} \sim \mathcal{N}(0, \sigma_0^2)$, $\mathbf{b}^{(L)} = \mathbf{0}$ ($\sigma_0 = 10^{-4}$), all corrections are $O(\sigma_0)$ (the inertia matrix $O(\sigma_0)$, hence its torque contribution $O(\sigma_0)$), so $\hat{\boldsymbol{\tau}} = \boldsymbol{\tau}_{\text{phys}} + O(\sigma_0) \approx \boldsymbol{\tau}_{\text{phys}}$ at initialisation.*

The model therefore *starts* at the validated analytical solution and training can only refine it; there is no cold-start regression of the rigid-body structure. The exactly-zero alternative is rejected because it deadlocks the $LL^\top$ inertia branch (Proposition 3); $\sigma_0 = 10^{-4}$ gives correction magnitudes negligible against the $O(1)$ physics baseline while keeping gradients alive from the first step.

10 Structural Guarantees (Summary)

For *every* parameter configuration, by construction:

1. $\hat{\boldsymbol{\tau}} = \boldsymbol{\tau}_{\text{phys}}$ to $O(10^{-4})$ at init (Prop. 6).
2. $\delta M = \delta M^\top$ (Prop. 2); in $LL^\top$ mode $\tilde{M} \succ 0$.
3. $\delta C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} \rightarrow \mathbf{0}$ as $\dot{\mathbf{q}} \rightarrow \mathbf{0}$ (all three Coriolis modes).
4. $\delta \mathbf{g}$ is independent of $\dot{\mathbf{q}}, \ddot{\mathbf{q}}$ (gravity stays T -even).
5. $\delta \boldsymbol{\tau}_f(-\dot{\mathbf{q}}) = -\delta \boldsymbol{\tau}_f(\dot{\mathbf{q}})$ (Prop. 5; both friction forms).
6. $\dot{\tilde{M}} - 2\tilde{C}$ skew-symmetric in the structural Coriolis mode (Prop. 4).

11 Training Objective and Optimisation

11.1 Composite loss

EDR minimises a data term, a per-component correction-magnitude regulariser, and an optional stability term:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \sum_{x \in \{g, M, C, f\}} \lambda_x \mathcal{L}_{\text{corr}}^{(x)} + \lambda_{\text{stab}} \mathcal{L}_{\text{stab}}. \quad (22)$$

Data term. A per-joint weighted error in normalised space (matching the trajectory macro-RMSE used for early stopping),

$$\mathcal{L}_{\text{data}} = \frac{1}{Bn} \sum_{b=1}^B \sum_{i=1}^n w_i (\tilde{\tau}_{b,i} - \tilde{\tau}_{b,i})^2, \quad \frac{1}{n} \sum_i w_i = 1, \quad (23)$$

with mean-normalised per-joint weights w_i (uniform $w_i = 1$ in the reference model; a Huber variant with $\delta = 0.5$ is available for outlier robustness).

Correction regulariser (Occam’s razor). Each correction is penalised so the model prefers nominal physics unless the data strongly disagree:

$$\mathcal{L}_{\text{corr}}^{(g)} = \overline{\|\delta\mathbf{g}\|^2}, \quad \mathcal{L}_{\text{corr}}^{(M)} = \frac{\overline{\|\delta\mathbf{M}\|_F^2}}{n}, \quad \mathcal{L}_{\text{corr}}^{(C)} = \overline{\|\delta\mathbf{C}\dot{\mathbf{q}}\|^2}, \quad \mathcal{L}_{\text{corr}}^{(f)} = \overline{\|\delta\boldsymbol{\tau}_f\|^2}, \quad (24)$$

where $\overline{(\cdot)}$ is the batch mean and the Frobenius inertia term is scaled by $1/n$ for per-element parity with the vector terms. Per-component weights λ_x are essential: the $O(n^2)$ matrix corrections $\delta\mathbf{M}, \delta\mathbf{C}$ have far more effective capacity than the $O(n)$ vectors $\delta\mathbf{g}, \delta\boldsymbol{\tau}_f$ and require stronger penalties, lest they memorise; the weights may optionally be cosine-decayed over training (strong Occam early, relaxed late).

Stability term (optional). A soft skew-symmetry penalty enforcing Property 5 when forms (a)/(b) are used,

$$\mathcal{L}_{\text{stab}} = \overline{\|S + S^\top\|_F^2}, \quad S = \dot{\tilde{M}} - 2\tilde{C}, \quad (25)$$

computed on a random subminibatch (the Jacobian $\dot{\tilde{M}}$ is moderately expensive). It is redundant — and disabled — under the structural Coriolis mode, where passivity already holds exactly (Prop. 4).

11.2 Optimisation and regularisation strategy

A single-group AdamW optimiser (the smooth gate removes the need for the two parameter groups of the two-phase scheme) is used with a warm-up+cosine learning-rate schedule (cosine annealing with warm restarts in the original setup), gradient-norm clipping, and **float32 throughout** — mixed precision is disabled because the structural-Coriolis Jacobian is unreliable in float16. Generalisation rests on several mild, complementary regularisers acting at different layers: weight decay, the correction-magnitude penalty (24), dropout in the δ -net hidden layers, small input feature-noise augmentation, and early stopping on the validation macro-RMSE; an exponential moving average (EMA) of the weights may be maintained for model selection. The design philosophy is that several gentle regularisers keep the training and validation curves descending in lockstep, avoiding the post-peak overfitting a single strong penalty would invite.

12 Reference Configuration and Parameter Budget

Table 2 lists the architectural hyper-parameters of the EDR model used in the study, and Table 3 its per-component parameter budget; these are architectural facts, not results. The budget scales with the hidden widths — a smaller $[32, 32]/[16, 16]$ variant totals $\approx 5.4\text{k}$ parameters — but every parameter is structurally constrained.

13 Benefits over the Baselines

Versus the black-box network. The black box must rediscover the entire rigid-body structure from data: SPD inertia, velocity-quadratic Coriolis, configuration-only gravity, and odd friction are nowhere encoded and hold, at best, approximately and only on the training distribution. EDR encodes all of these exactly and for free, spending its few parameters on the small residual rather than on relearning physics. This yields far fewer parameters, a strong inductive bias for data efficiency and generalisation, physically plausible off-distribution behaviour (the analytical

Setting	Value	Setting	Value
Joints n	5	Activation ϕ	SiLU
$\delta\mathbf{g}/\delta\mathbf{M}/\delta\mathbf{C}$ hidden	[48, 48]	Friction hidden	[24, 24]
Trig features (sin / cos)	enabled	Physics conditioning	enabled
Inertia mode	symmetric scatter	Coriolis mode	element-wise
Friction form	MLP (+ $ \ddot{\mathbf{q}} $)	Final-layer init σ_0	10^{-4}
Correction dropout p	0.30	Gate ramp fraction ρ	0.30
Optimiser	AdamW (single group)	LR schedule	warm-up cosine
Grad-clip norm	1.0	Precision	float32
Weight EMA	enabled	Early stop metric	val macro-RMSE

Table 2: Reference EDR architecture configuration.

Network	Output / structure	Input dim	Parameters
Gravity $\delta\mathbf{g}(\mathbf{q})$	$\mathbb{R}^n$, free MLP	20	3,605
Inertia $\delta\mathbf{M}(\mathbf{q})$	$m = n(n+1)/2 = 15$ entries $\rightarrow \text{Sym}(n)$	20	4,095
Coriolis $\delta\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$	$\dot{\mathbf{q}} \odot \text{MLP}_C$ (element-wise)	40	4,565
Friction $\delta\boldsymbol{\tau}_f(\dot{\mathbf{q}})$	$\dot{\mathbf{q}} \odot h_\phi(\dot{\mathbf{q}} , \ddot{\mathbf{q}} , \boldsymbol{\tau}_f)$	15	1,109
Total			13,374

Table 3: Per-component parameter budget of the reference EDR model.

backbone dominates and the correction is bounded), and component-level interpretability a monolithic MLP cannot offer.

Versus the physics-regularised network. The physics-regularised model *feeds* the analytical decomposition to an unstructured MLP and/or penalises departure from $\boldsymbol{\tau}_{\text{phys}}$; physics is a hint, not a constraint, so nothing prevents the learned function from violating symmetry, oddness, or velocity-vanishing, and the model stays large because it carries a full input-to-output regressor. EDR makes physics *architectural*:

- **Exact structure, not soft hints** — symmetry, oddness, velocity-vanishing, and (structural mode) passivity hold for every weight and cannot be traded away to lower the training loss (Props. 2, 4, 5).
- **Parameter efficiency** (Table 4) — the corrections are tiny because each models only a small structured residual; the analytical terms carry the bulk of the prediction at zero parameters.
- **Interpretability** — the four correction magnitudes ($\|\delta\mathbf{g}\|$, $\|\delta\mathbf{M}\|_F$, $\|\delta\mathbf{C}\dot{\mathbf{q}}\|$, $\|\delta\boldsymbol{\tau}_f\|$) reveal which physical term the URDF/friction model gets wrong and by how much.
- **Safe, monotone refinement** — zero-init plus the Occam regulariser means EDR starts at the validated physics model and departs only where the data demand.
- **Deployability** — in the structural mode $(\tilde{M}, \tilde{C})$ remain a valid, passive rigid-body pair, suitable for model-based control with stability guarantees.

14 Empirical Evaluation

A quantitative comparison of EDR against the black-box and physics-regularised baselines — accuracy, parameter and data efficiency, generalisation, and per-joint behaviour — is reported

Model	Trainable parameters	Physics use
Black-box FNN	$\sim 70\text{k}$	none
Physics-regularised FNN	$\sim 549\text{k}$	additive (input/penalty)
EDR (this work)	$\sim 13\text{k}$	structural (architectural)

Table 4: Parameter/physics comparison of the three modelling paradigms (orders of magnitude; EDR is $\sim 5\times$ smaller than the black box and $\sim 41\times$ smaller than the physics-regularised baseline).

separately and is intentionally omitted here.¹

15 Conclusion

EDR carries physics from the *input* and *loss* of a learner into its *structure*. Each analytical torque component is refined by a tiny network that is equivariant to the component’s symmetry, near-zero at initialisation, and penalised toward silence. The manipulator equation’s exact symmetries — inertia symmetry, configuration-only gravity, velocity-vanishing quadratic Coriolis, odd friction, and passivity — are preserved by construction rather than merely encouraged, giving a compact, interpretable, and physically consistent model whose inductive bias is exact.

¹Results, tables, and figures to be inserted by the author.